Back to Basics - Strategic Report

A strategic report for Back to Basics, an off-grid urban hemp farm in Hackney Wick. It sets out the legislative framework, project management and procurement strategy needed to deliver a low-energy, community-led development that grows, processes and builds with hemp.



The report structures a strategy for team procurement, funding and phased delivery, informed by planning and building regulations, stakeholder engagement and risk assessment, to bring the Back to Basics hemp farm and housing scheme to reality.

01/01/2023 - date started

15/03/2023 - date ended

Documents by Sebastian Dawber
https://www.sebastianvictordawber.co

Created for ...
eg BSc Architecuture and enviromental design
Undergraduate RIBA ARB part I

All work produced by Sebastian Victor Dawber

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https://www.sebastianvictordawber.co/

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Https://www.instagram.com/sarc_studio/



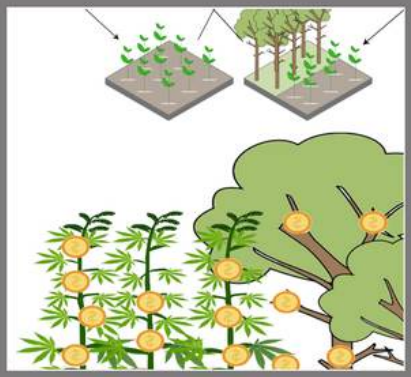
SEBASTIAN VICTOR DAWBER
Independent Project
BSc Part 1, ARB RIBA , Architecture and environmental design

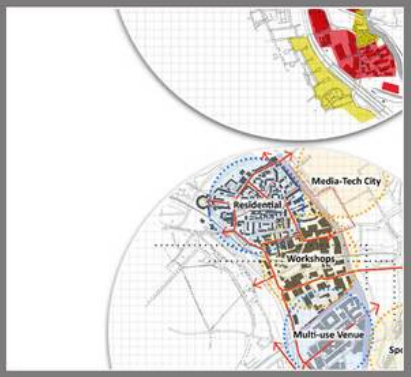
Back to Basics - Strategic Report

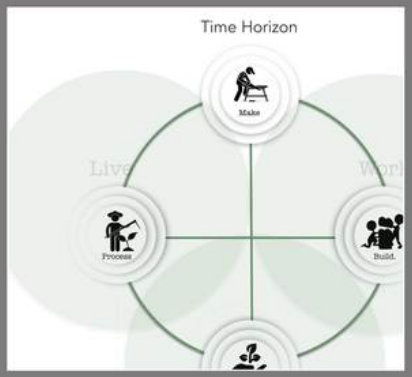
Procurement, Funding & Legislative Strategy

This document is the strategic report supporting Back to Basics, an off-grid urban hemp farm and housing scheme for Hackney Wick, Fish Island. As new development displaces the area's existing community, Back to Basics proposes a grow-process-make-build model that brings new and long-standing residents together around a shared agricultural landmark, offering cafe, growing, workspace, learning and exhibition spaces alongside hempcrete self-build housing. This report sets out the legislative framework, covering planning and building regulations, project management, procurement team and stakeholders, and funding, costs and risk, required to deliver the scheme. It concludes with a strategy to procure a team and funding, using passive and active strategies to lower energy demand and create a circular economy in which all parties benefit.

15/03/2023 - date ended

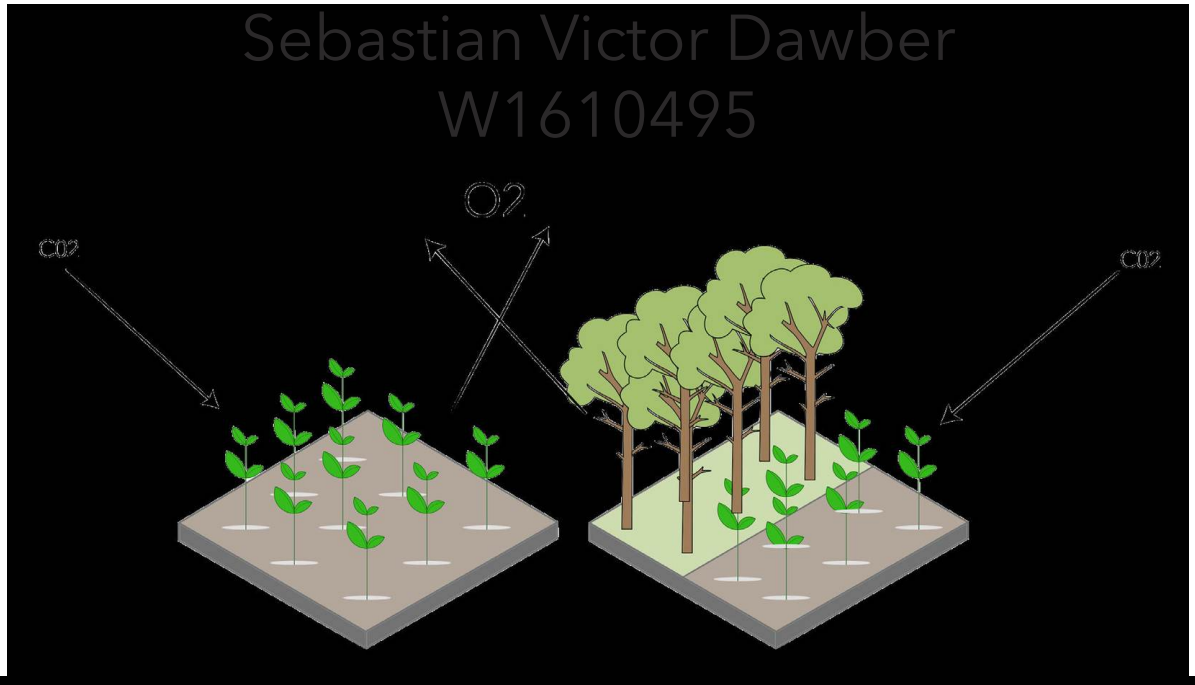






Startegic report

Sebastian Victor Dawber
W1610495





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Introduction

Back to Basics

Back to times of the past. A near to off-grid urban village that operates for and with the people of Hackney Wick. As developers move into Hackney Wick the original residents of this hamlet are slowly disappearing as a new community in Fish Island is emerging. The local community who are seeking jobs and homes are having to leave their beloved land, leading to a division between old and newcomers.

The idea is to create a community district that works with rather than fighting with new and old residents, a place of commerce for all—an urban community that has public and private access, welcoming all to the site. To grow the community; spaces such as a café, crop growing, workspaces, learning rooms, exhibition spaces and social housing— all aimed to increase cohesion amongst the community. Further explored in a chance to work on-site as well as partake in a hempcrete self-build project. People can live on-site and grow their own spaces using hempcrete, using the hemp plant to draw people in through curiosity and wow them with a modern green building design, using architecture to settle the un-settled- an urban farming settlement.

The site services use traditional and modern strategies to create an off-grid industrial processing space. Using little or no services the building does not harm or fight the community; rather, it fights to conjure them into one magical potion that can be used to cure the division in society. A village to grow the community connecting to opposing sides into one wholesome community.

Design statement

Back to basics

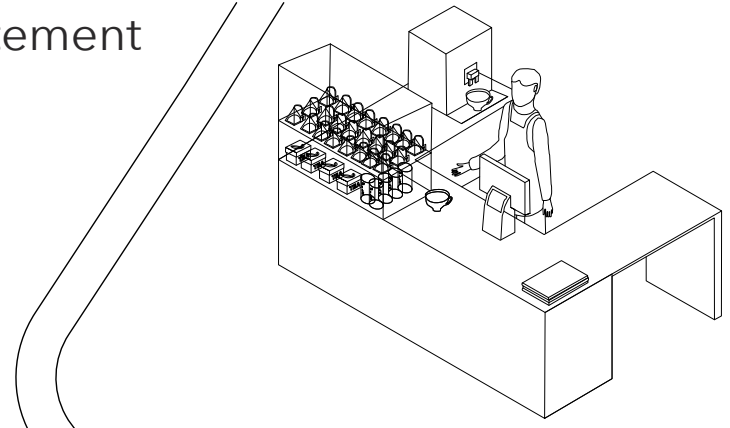
Project brief

G- grow, P- process ,M- make ,B- Build, S- Sell

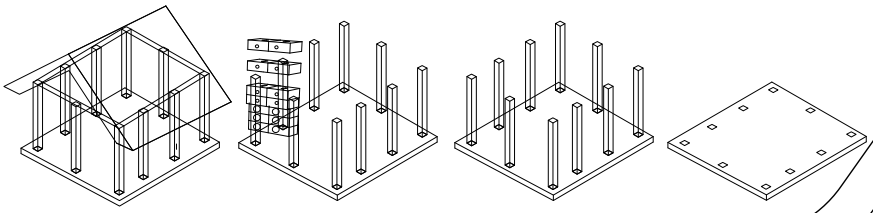
- Harvests occur every 2 months

- 1 hectare (10,000.sqm) produces enough hemp for a 120 sqm house

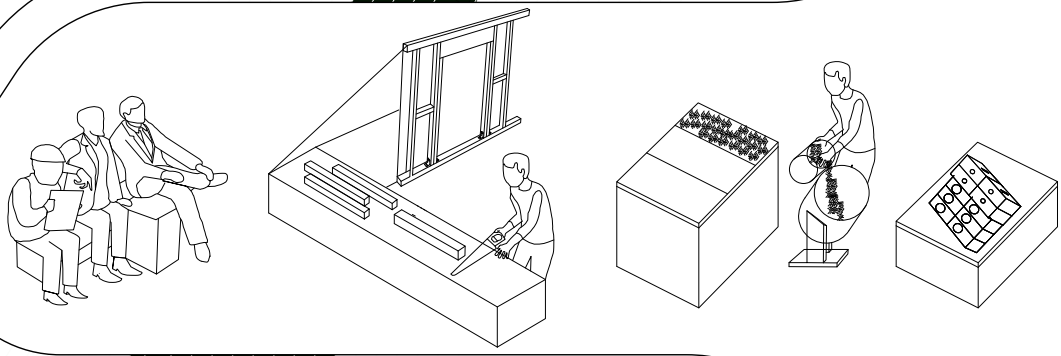
My design has 5 596 sqm of growing space which produces enough hemp for a 120 sqm house every 4 months, thus Back to basics produces enough hemp for 3 houses at 32sqm every 4 months with 4 houses being produced every year those that work on site are able to grow and build their own house on site to create a circular community overtime



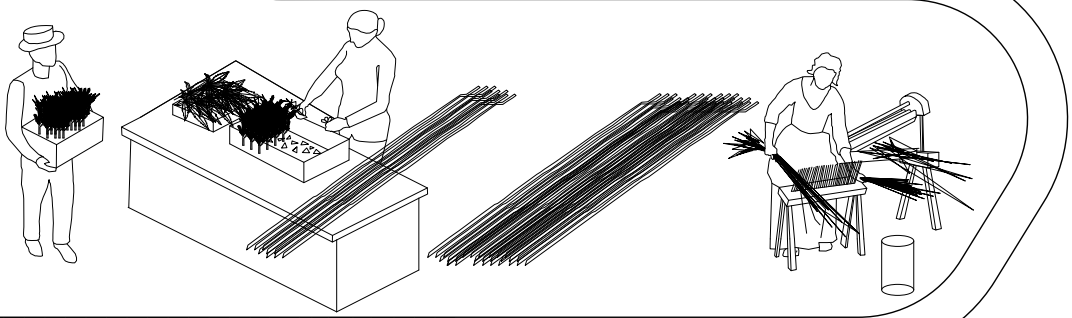
8am to 5pm Sell



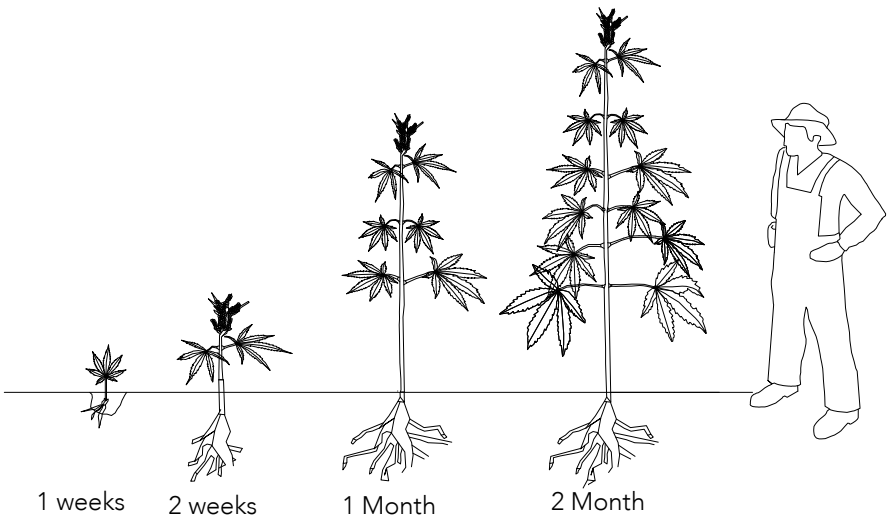
8 am to 5pm Build



12 am to 2pm Process, make, learn



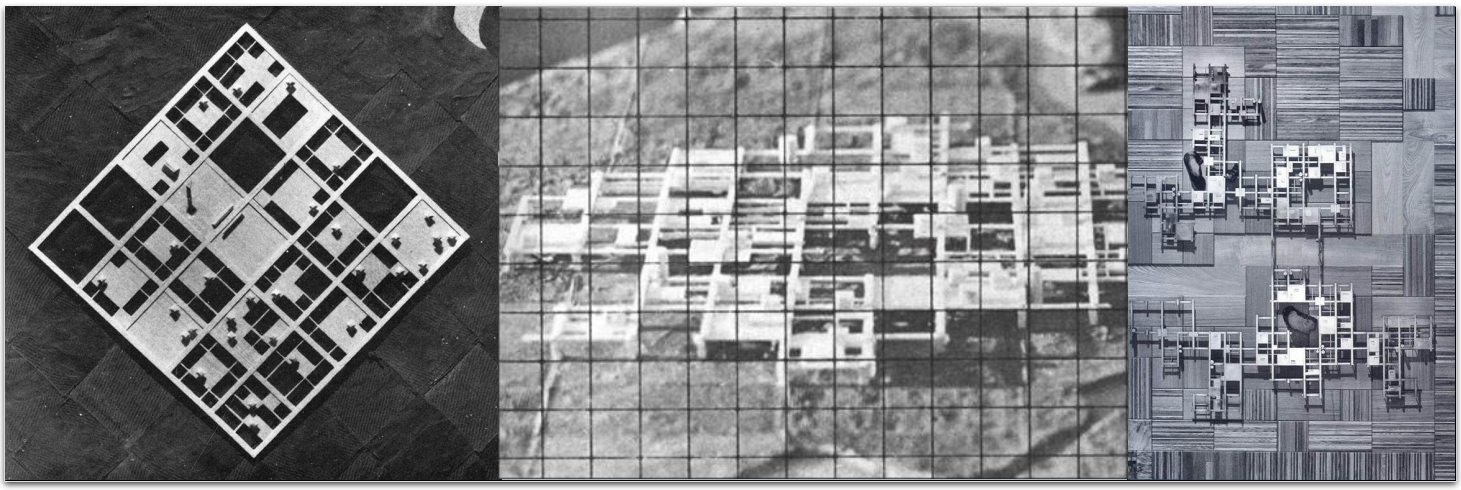
24hrs hrs 24hrs Grow

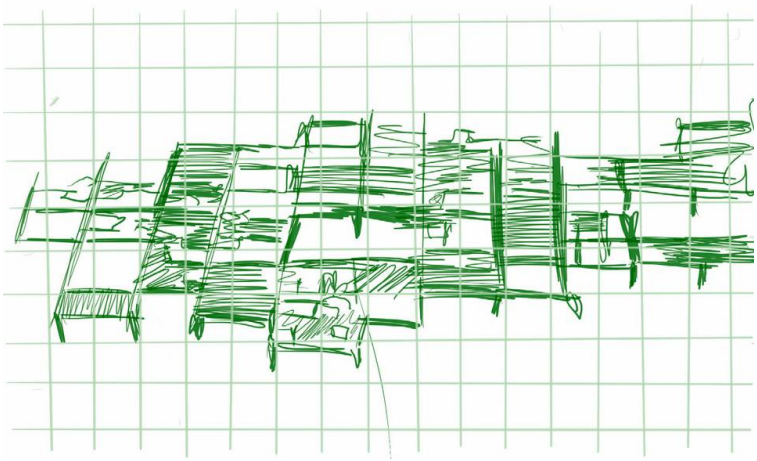


Precedents

Key research

Agricultural City, 1960 / Kisho Kurokawa

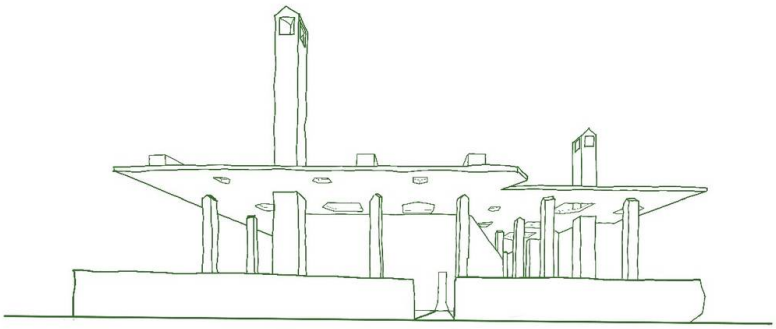




The natural growth of the agricultural city is provided by a grid system of streets containing the utility pipes underneath. While each of the square units composed of several households is autonomous, linking these units together creates a village. The living units multiply spontaneously without any hierarchy, gradually bringing the village into being as the traditional rural settlement has developed throughout Japanese history. The units are lifted off the ground due to flooding and earthquake risks allowing farming to take place on the ground level.

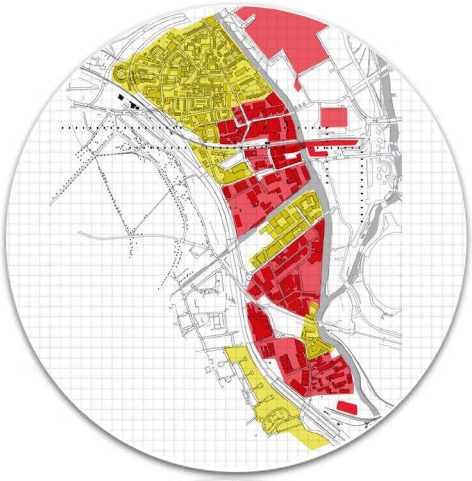
Pearling Site Museum and Entrance / Valerio Olgiati





The site contains ruins that form part of the UNESCO Pearling Path. The entire building functions as the entrance to the cultural heritage and the foyer for the medina. It is an urban room for the people of Muharraq with the scale of a public park. Concrete elements are placed along the property boundary to form a new locus in the dense city. As a totality, the building creates a universe in itself that is the entrance for the Pearling Path and the city beyond.

Key research

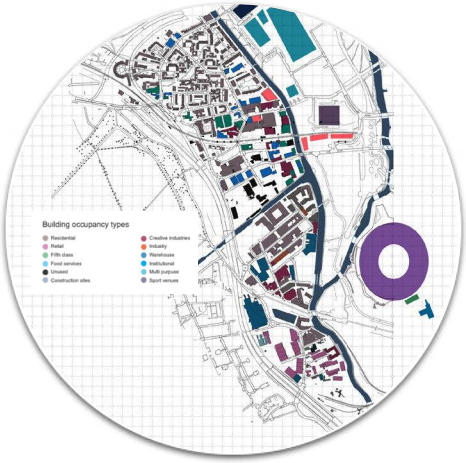


An understanding of the site was formed through a macro analysis to understand the area. It allowed me to see the type of spaces one can currently find and how my design can fit similar trends to continue was is already in place rather than remove its legacy. The start of my functions started to form.



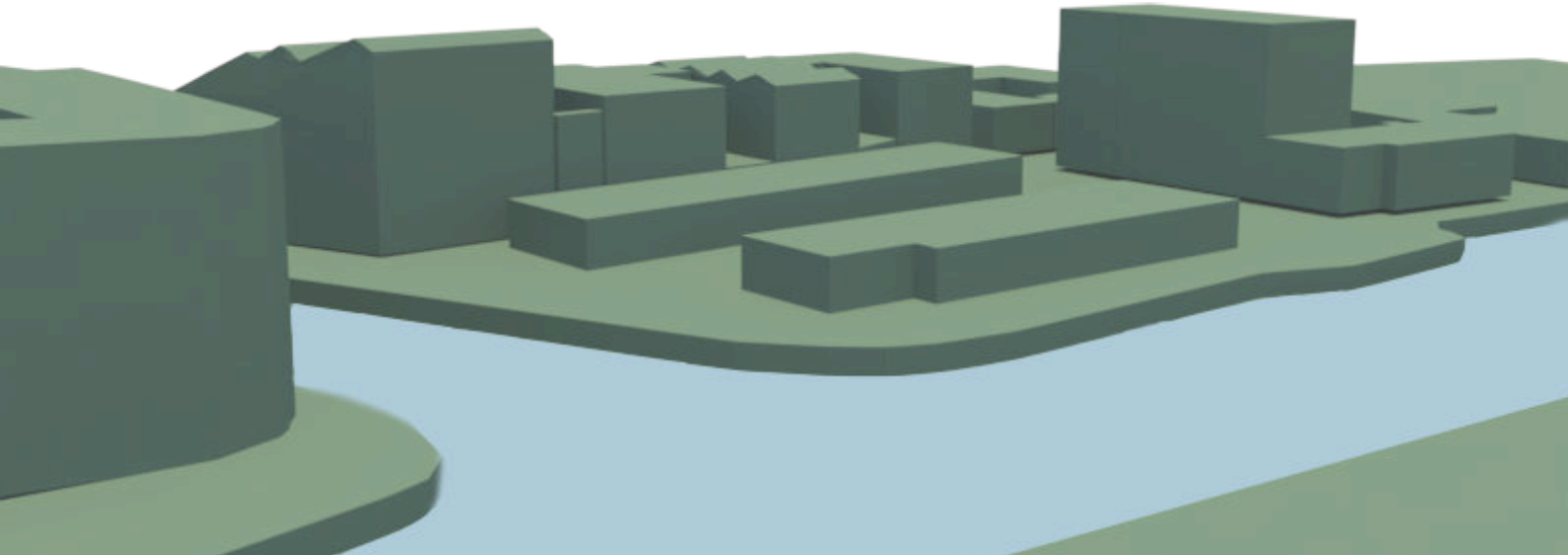
The history of the site showed a change from an industrial area marked in red to a residential area marked in red. This change means people have to move out of the area and find new places of work.

Different typologies revealed various uses. The idea of a multi-use workshop venue appealed due to human first-hand research as well as the idea to create an urban village. An industrial space to serve work, social meetings and homes.



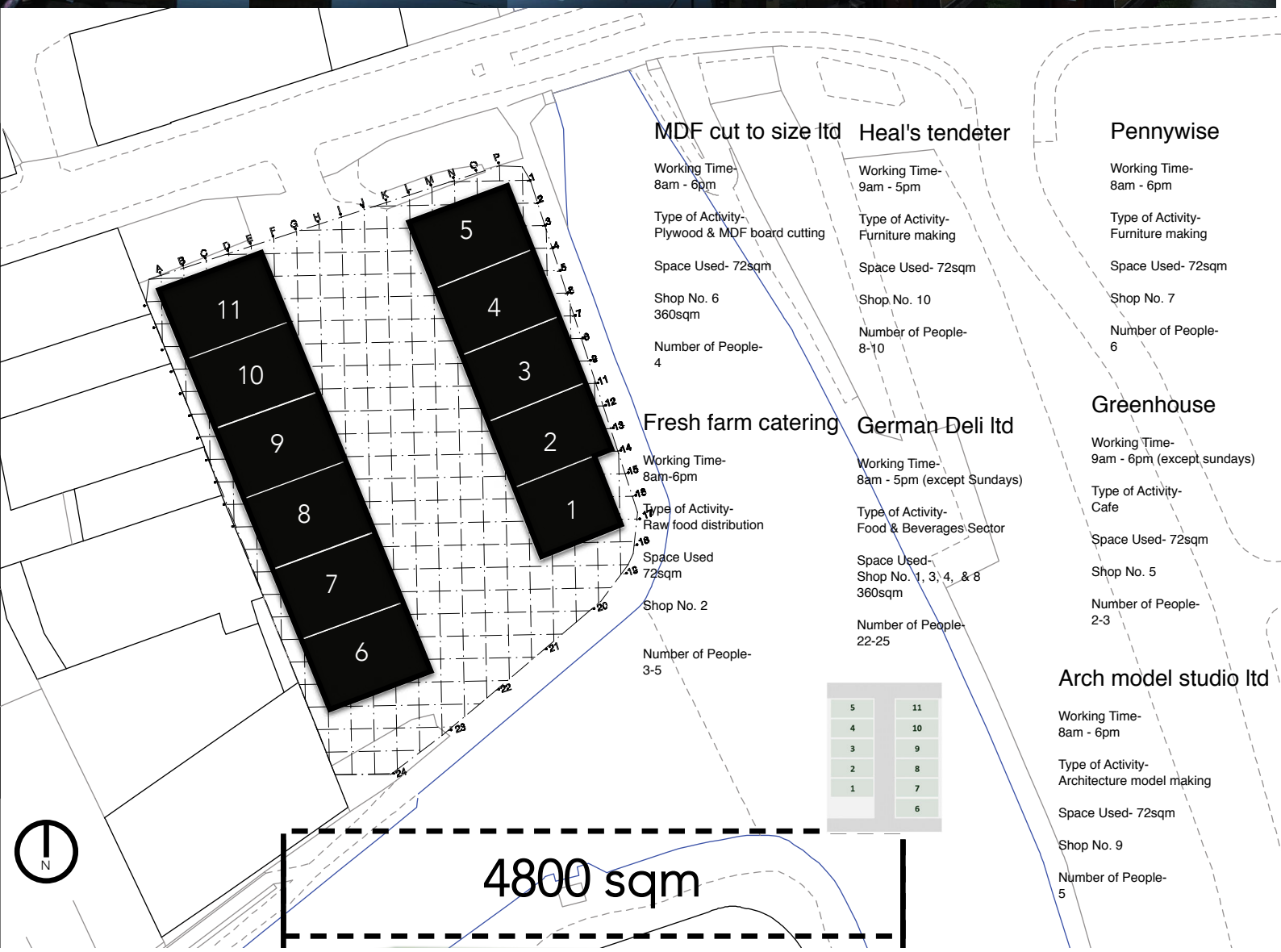
More in-depth research into the buildings revealed a multitude of uses. Warehouses, unused, multipurpose spaces appealed due to the intent of the design. A multi-use venue to maximise space used to grow crops while providing for all of those in the area.

This macro analysis of the entire site area, including Hackney Wick and Fish Island, showed similarities in the area and how they could be combined into one wholesome community through design.



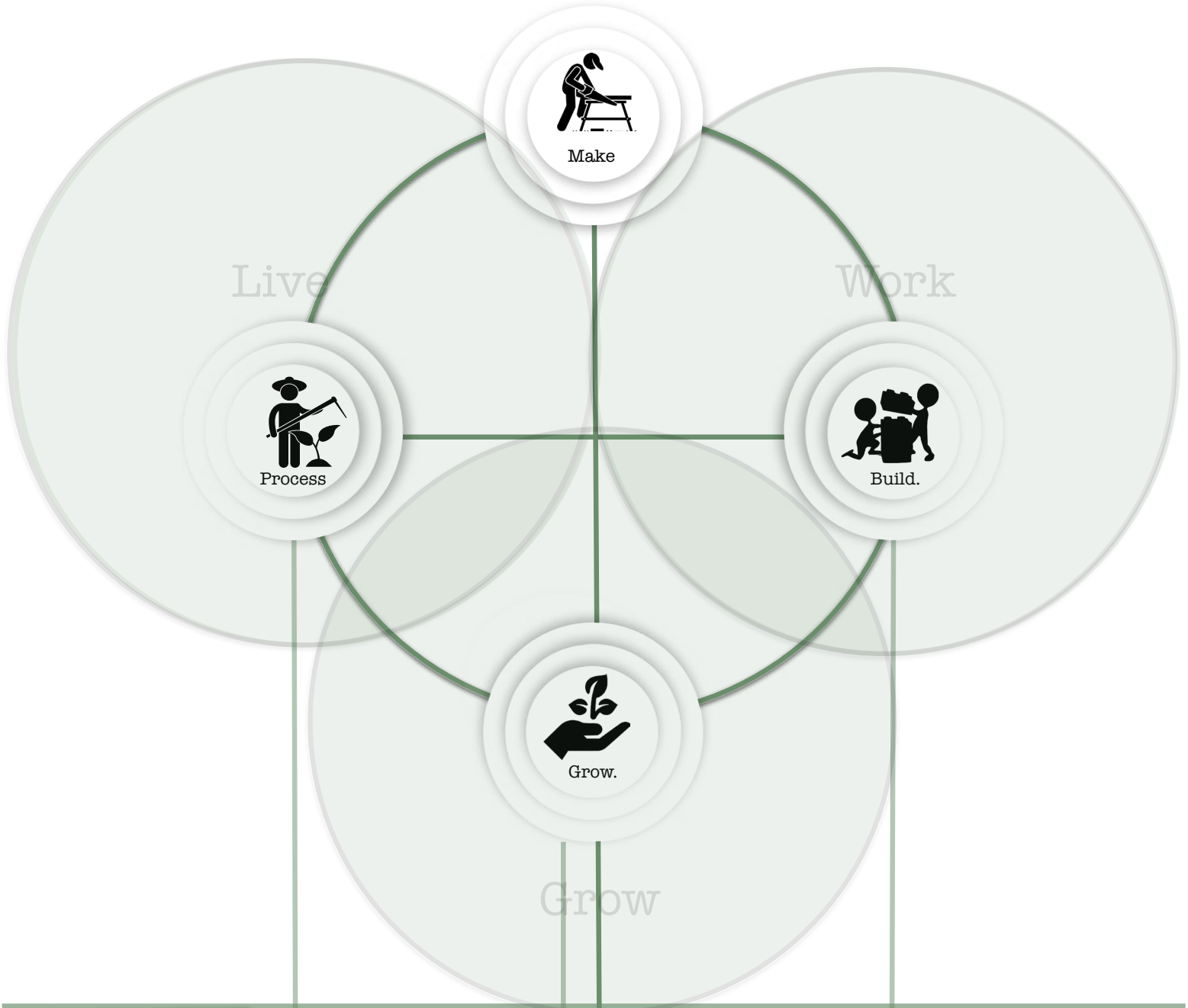
Key research





The macro site analysis was the start of my design functions. To understand the chosen site and to ensure the same people aren't displaced. I integrated the existing site functions into my design With the macro analysis in mind, it formed the final steps of my site functions. The existing site functions were workshop services, carpentry services, architectural studios, fresh food catering, and café. My design intends to keep the following: café, workshop services, carpentry services, architectural studios and offer jobs to those who worked in the fresh food catering.

Time Horizon



Every morning at 8am the farmer wakes up

Between 8am- 17pm plant processing takes place

-Breaking
-Striping
-Sorting
-Drying

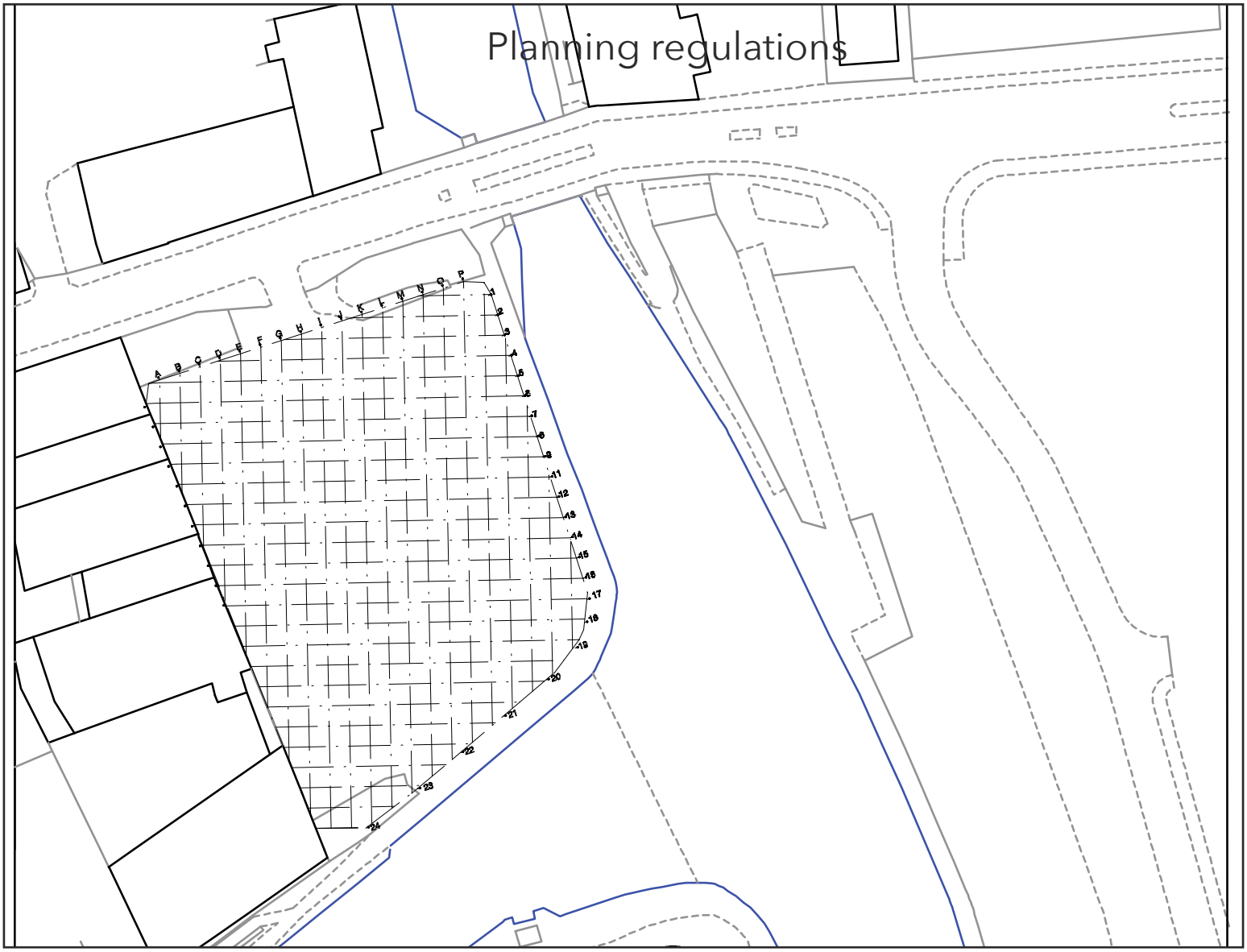
Hempcrete is made between 10am-4pm where the public can view the process from the central courtyard.

Once the hemp is grown and the materials are made the building process takes place between 9am-4pm

At end the day the system is turned off and plants are inspected to assess whether they have been damaged throughout the day during their photosynthesis period where they are most venerable.

8:00am 9:00am 10:00am 11:00am 12pm 12:30pm 13:00pm 14:00pm 17:00pm 18:00pm

Planning regulations



The idea of a multi-use space requires varied planning applications, as well as an application to HM, government to grow industrial hemp.
Applications such as: A3 for restaurants and cafes, B1 Business, C3 Dwelling houses and an Industrial Hemp (low THC Cannabis) cultivation licence. Once approved, the approved functions will be distributed on-site.

A3 will cover the café, which will serve tea and various ethic deserts to encourage people to come to taste their origins and talk about their ethnicity. Placed on the corner to maintain views the café will have an open space floor plan with sliding doors that allows the space to open during winter, creating an open-air café during the warmer months.

B1(c) (business - light industrial) the mix used space requires a B1(c) as the mixed-use spaces for industrial and business purposes. The workspaces will be closer to the road with car parking space while the industry space is closer to the river near the Fish Island entrance as it is the tool to draw people into the site.

C3 (dwelling houses) would be needed for the social housing scheme. The intent is to start with 10 homes then build to 20 overtime. This application is vital to ensure the overall project is given the go-ahead.

Finally, the second most crucial aspect after C3, Industrial Hemp (low THC Cannabis) cultivation licence which will allow the hemp growing to take place. Restrictions are applied to the application, and if they deny the application, the hemp will have to be sourced from UK hempcrete ©.

Building regulations

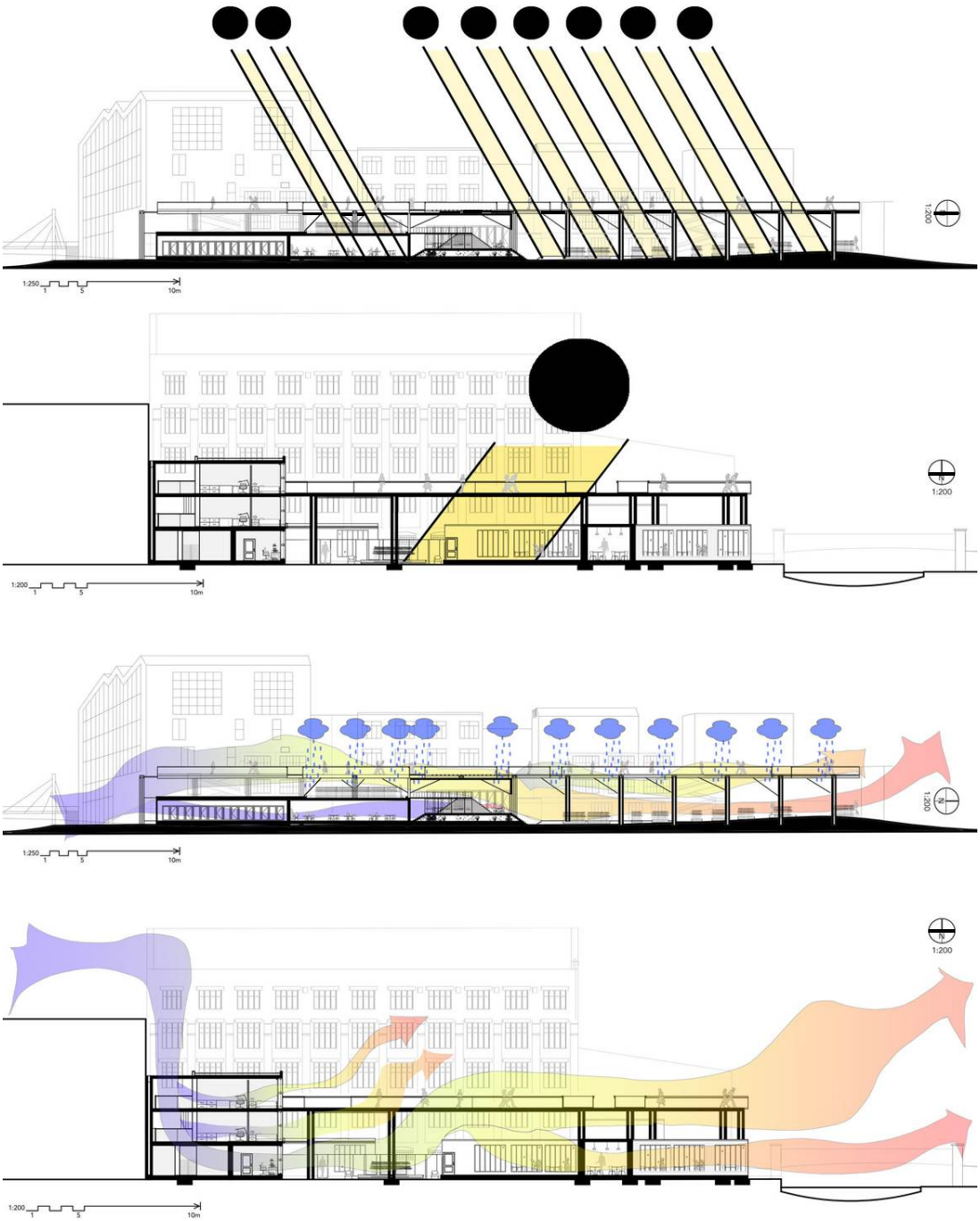


M the access to and use of buildings.

Referring to: Category 1 – visitable dwelling Access and use
M4(1). Reasonable provision should be made for people to–
(a) gain access to; and
(b) use, the dwelling and its facilities
Requirement M4(1) does not apply to:
(a) an extension to a dwelling; or
(b) any part of a building that is used solely to enable the building or any service or fitting in the building to be inspected, repaired or maintained.

The marked growing space has restrictions and must be separated from public view. The grow spaces will only allow access to people older than 18 and will have a shutter system that can be moved around the space to close it during grow seasons. The hemp growing areas will only have access to the public during tours and everyone on the tour must be older than 18.

Building regulations

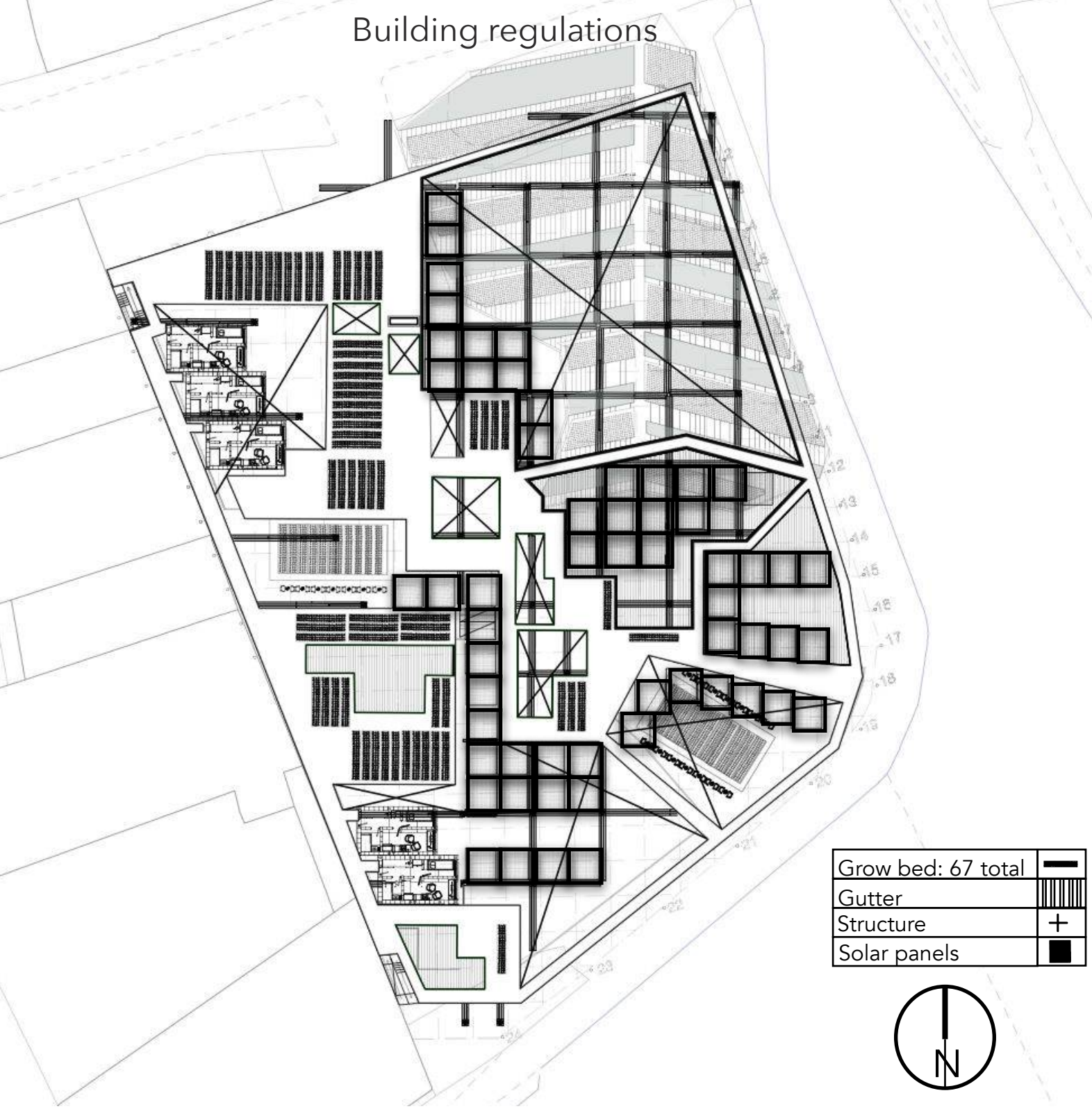


Part L Conservation of fuel and power

Schedule 1 – Part L Conservation of fuel and power L1. Reasonable provision shall be made for the conservation of fuel and power in buildings by:
(a) limiting heat gains and losses–
(i) through thermal elements and other parts of the building fabric; and
(ii) from pipes, ducts and vessels used for space heating, space cooling and hot water services; (b) providing fixed building services which– (i) are energy efficient;
(ii) have adequate controls; and
(iii) are commissioned by testing and adjusting as necessary to ensure they use no more fuel and power than is reasonable in the circumstances.

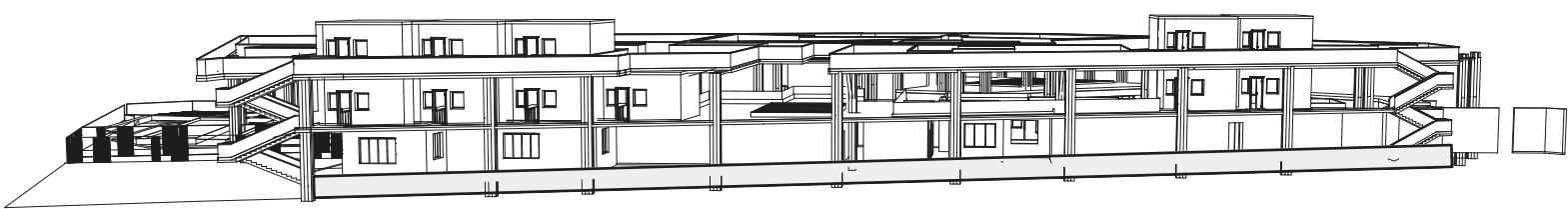
To conserve power, passive strategies were developed. The diagrams given show various performance such as cross-ventilation, stack ventilation, passive solar design and an adaptable system that can move with the space per season. The system collects water and allows for passive solar as well as thermal controls.

Building regulations



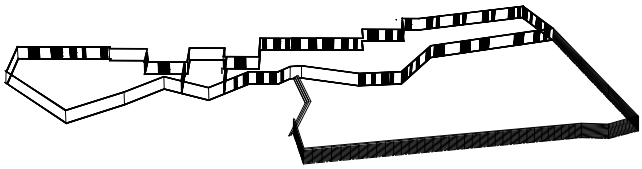
Further, solar panels and storable solar power Tesla walls are added to allow the building to run off-grid producing its own electricity. Each solar panel has 2,8kwh of energy with 60 in total so the site produces 168kwh per day. The Tesla Powerwall stores up to 13kwh. The energy that is produced but not used is kept and stored, during summer this is in excess thus in winter when days are dark power will be provided through stored energy
The Powerwall's are placed u

nder the stairs on the back wall marked in grey with a total of 150. The site stores enough power to power the site for 3 months running solely off stored energy in the power walls.



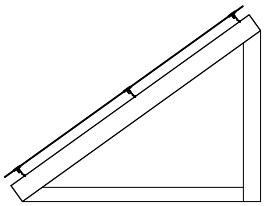
CDM regulations

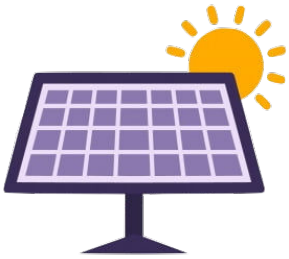
Constructive design regulations CDM follow the buildings regulations and are vital to ensure the safety and health of all people using the site. The principal design regulations for my design revolve around: security and safety of the hemp growing, instalment of the solar panels and the railing on the roof.



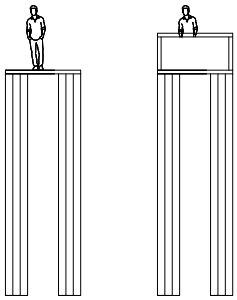


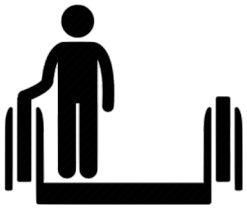
Shutters are added around the site growing areas. The rules stipulated within the industrial hemp growing license say that you need to keep the growing site disclosed away from public view. People above the age of 18 can view it but within a closed space. As the site welcomes all ages, the growing areas are separate to the sites café which is open to the public, with the drying phase viewable for exhibition purposes.



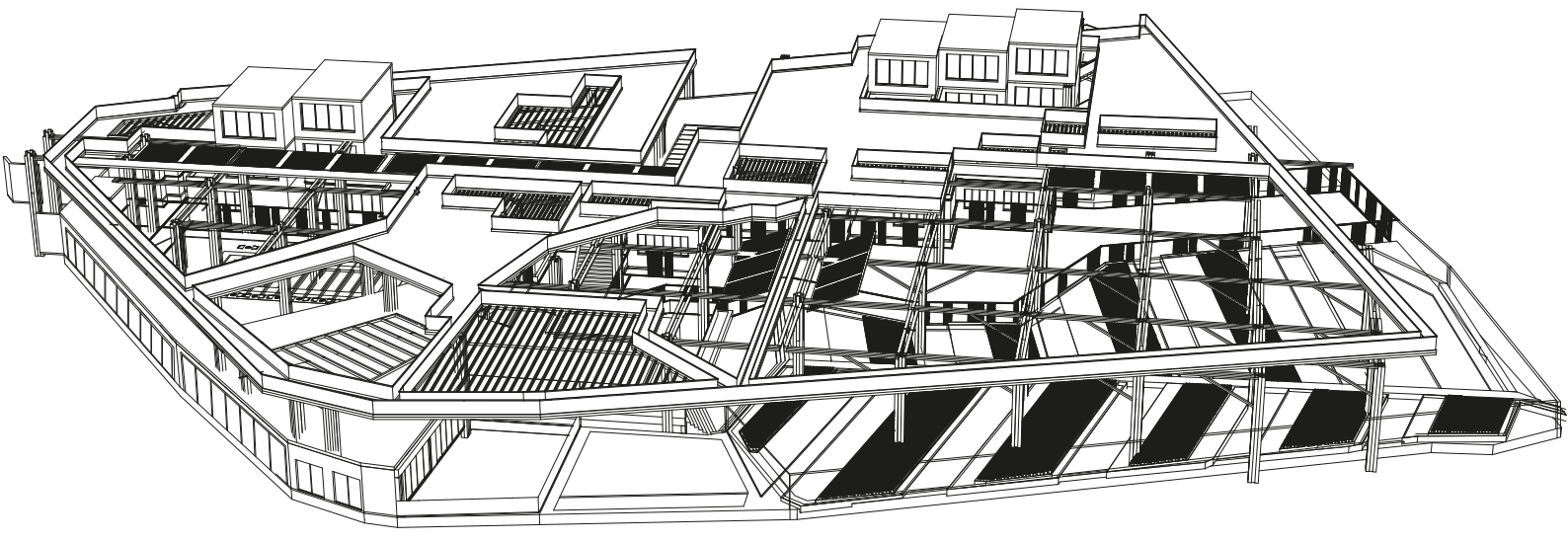


Solar panels are added when the building's structure is formed to ensure safety for those installing and to make sure the building is not damaged. Each panel can be accessed via the roof.





Lastly, rails are added to ensure people do not fall off the roof. This safety measure is essential as the 6m height may lead to serious injury upon no rails being present.



Team



Pam Alexander OBE

Director of the London Legacy Development Corporation (LLDC)



Valerio Olgiati
Architect



Duncan Walters
structural engineer



Bea Pernia
Interior desinger



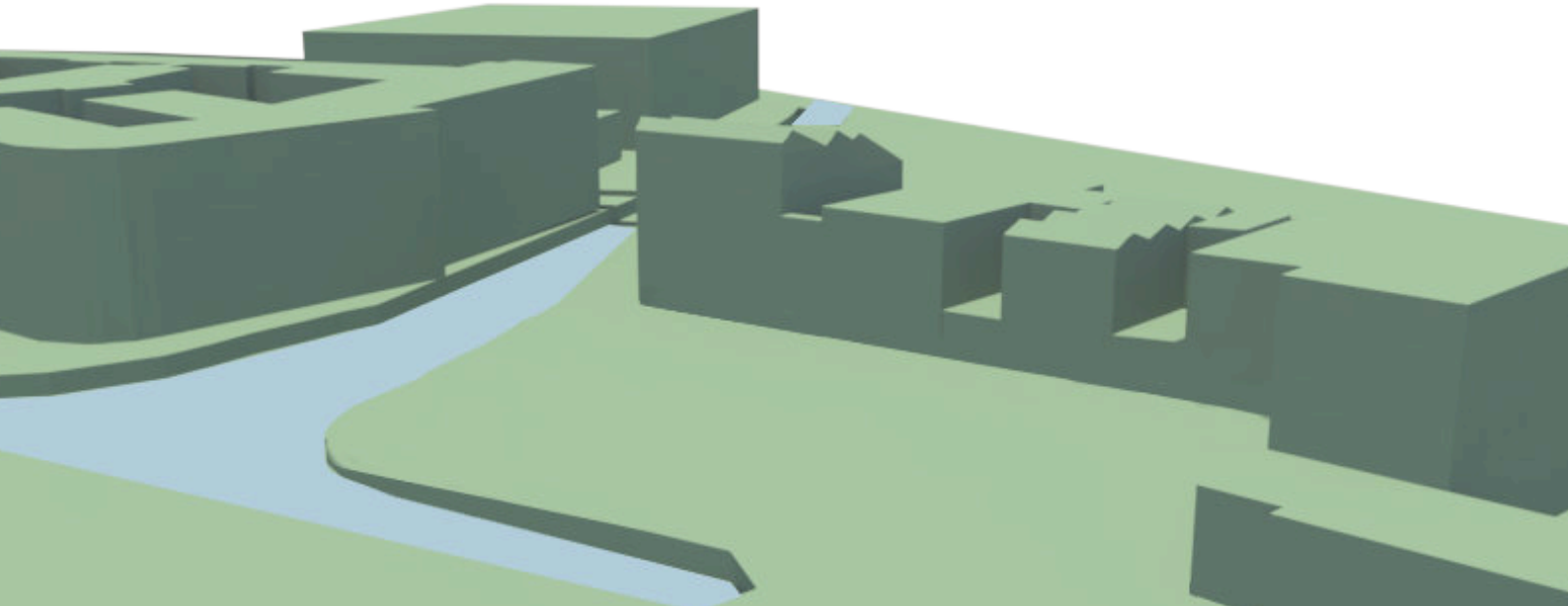
Jasmine Bedford
Environmental consultant



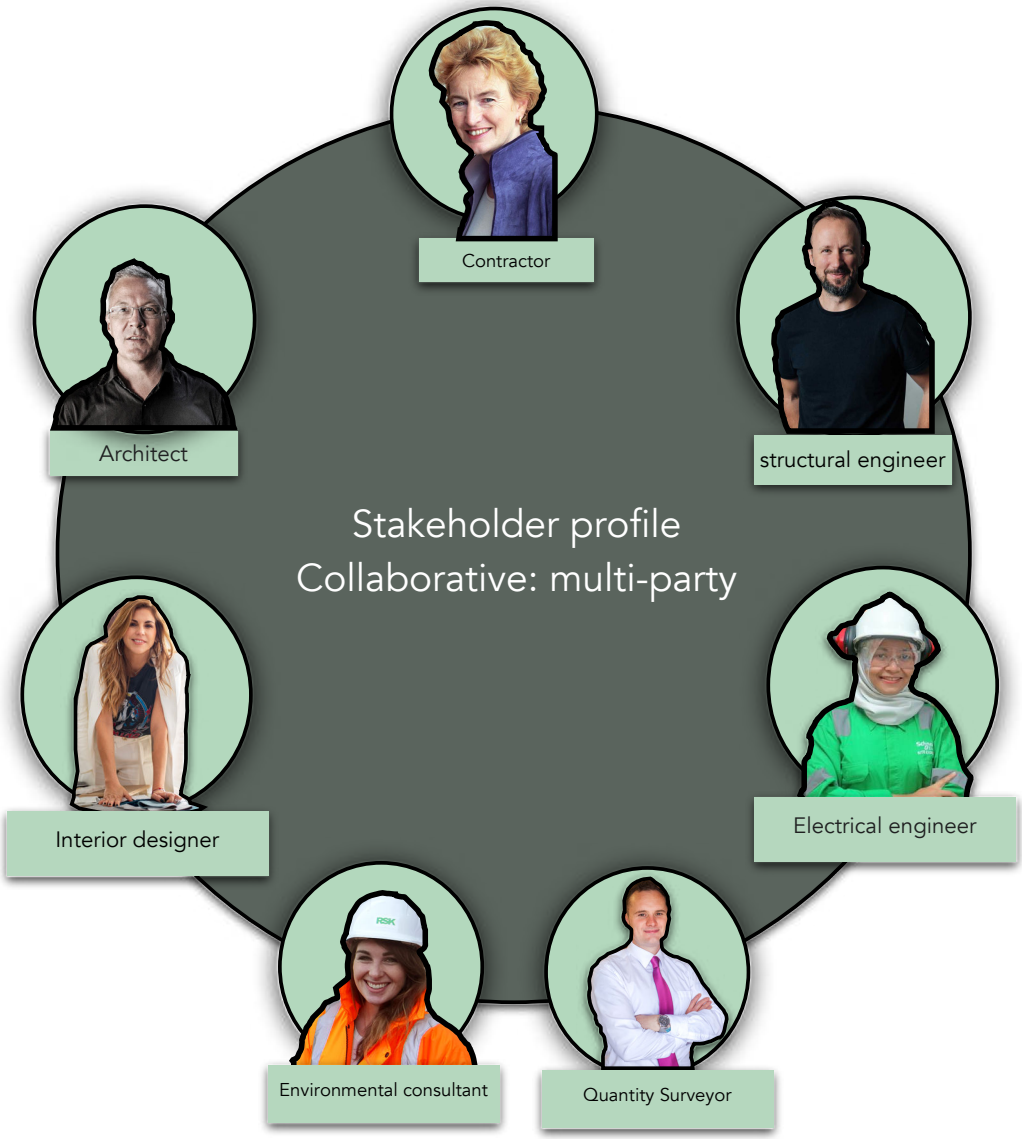
Ryan Claves
Quantity Surveyor



Maulidya Falah
Electrical engineer



Procurement team

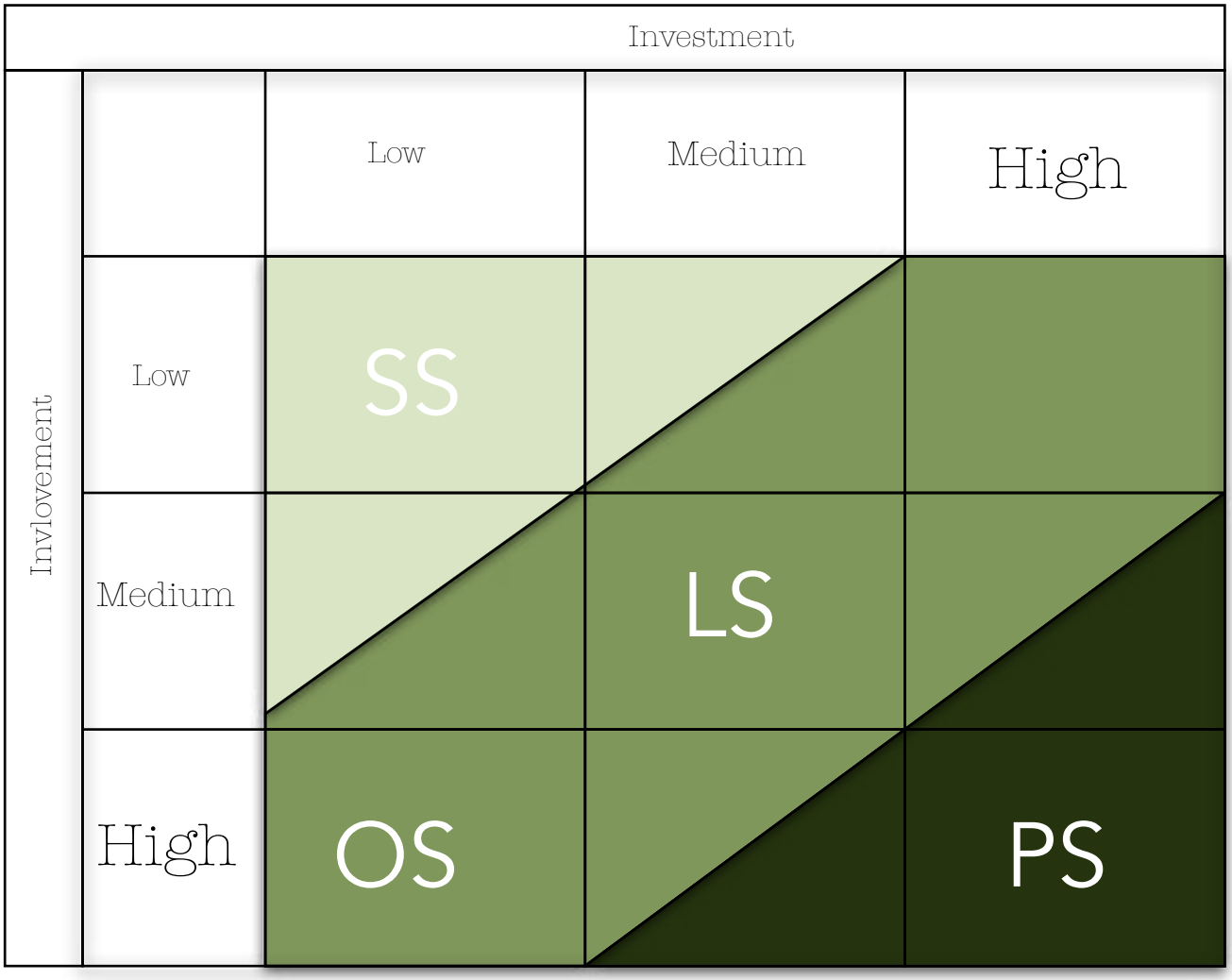


Procurement is vital to ensure the success of the project. To follow the design intent and to bring people together, a similar strategy will be implemented within the procurement process. A collaborative multi-party plan will be developed. Due to the following: Open book approach to risk premiums, risks are shared, 'Spirit of mutual trust and co-operation, requires good leadership of design and construction team and has the potential for 'Guaranteed maximum price' which will interest those funding the project.

The architect and engineer will work together to create the design as well as determine the cost. Once they have calculated the cost and finalised the design they will approach the contractor who will lead the procurement through government funding.

Pam Alexander will lead the procurement team in a collaborative approach working with all members involved. She will be key in bringing one together. The procurement will be the start of the idea of bringing people together, and a successful procurement may follow with success in the design, the first step to get people working and coming together.

Stakeholders



PS

Professional stakeholders

-Part 3 Architect
-ED Architect
-Interior Desinger

-Structural engineer
-Electrical engineer
-Quantity surveyor
-Finciaial Consultants

SS

Support stakeholders

-Hackney Wick locals

Back to Basics

LONDON LEGACY DEVELOPMENT CORPORATION

Client
LLDC

Social stakeholders

End users
- Visitors
-Consumers
-Students
-Workers
-Artists
-Architects
-Worksmen

LS

Legislative stakeholders

-Hackney London Borough Council
-Tower Hamlets Council
-Capri
-Parks and green spaces
-Sadiq Khan

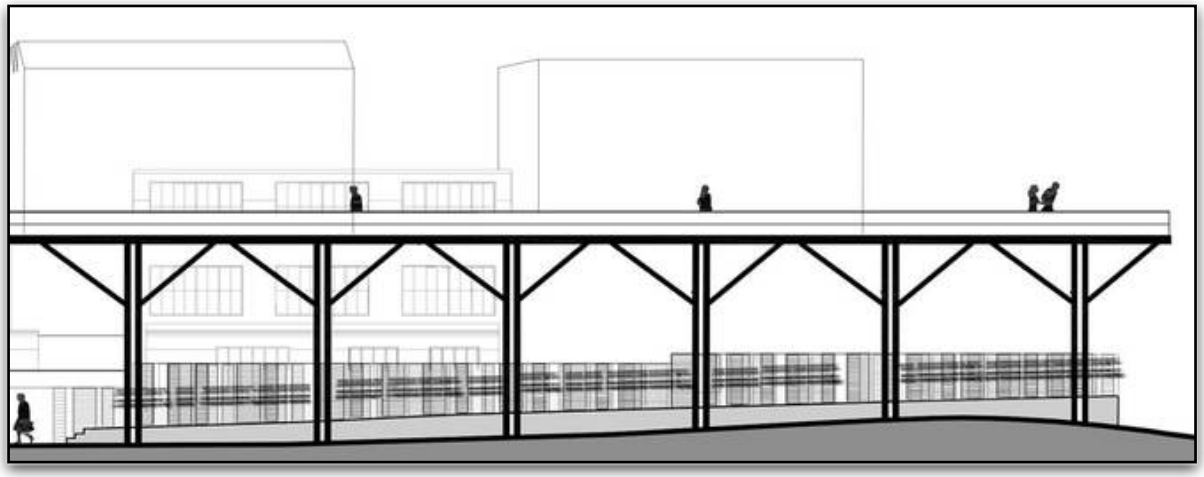
OS

Other indirect stakeholders

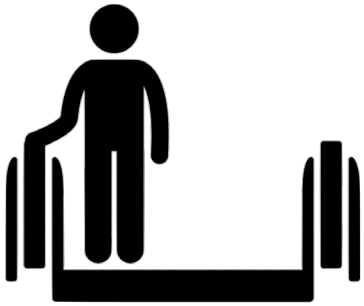
-HM government
-Metropolitan police

Risks





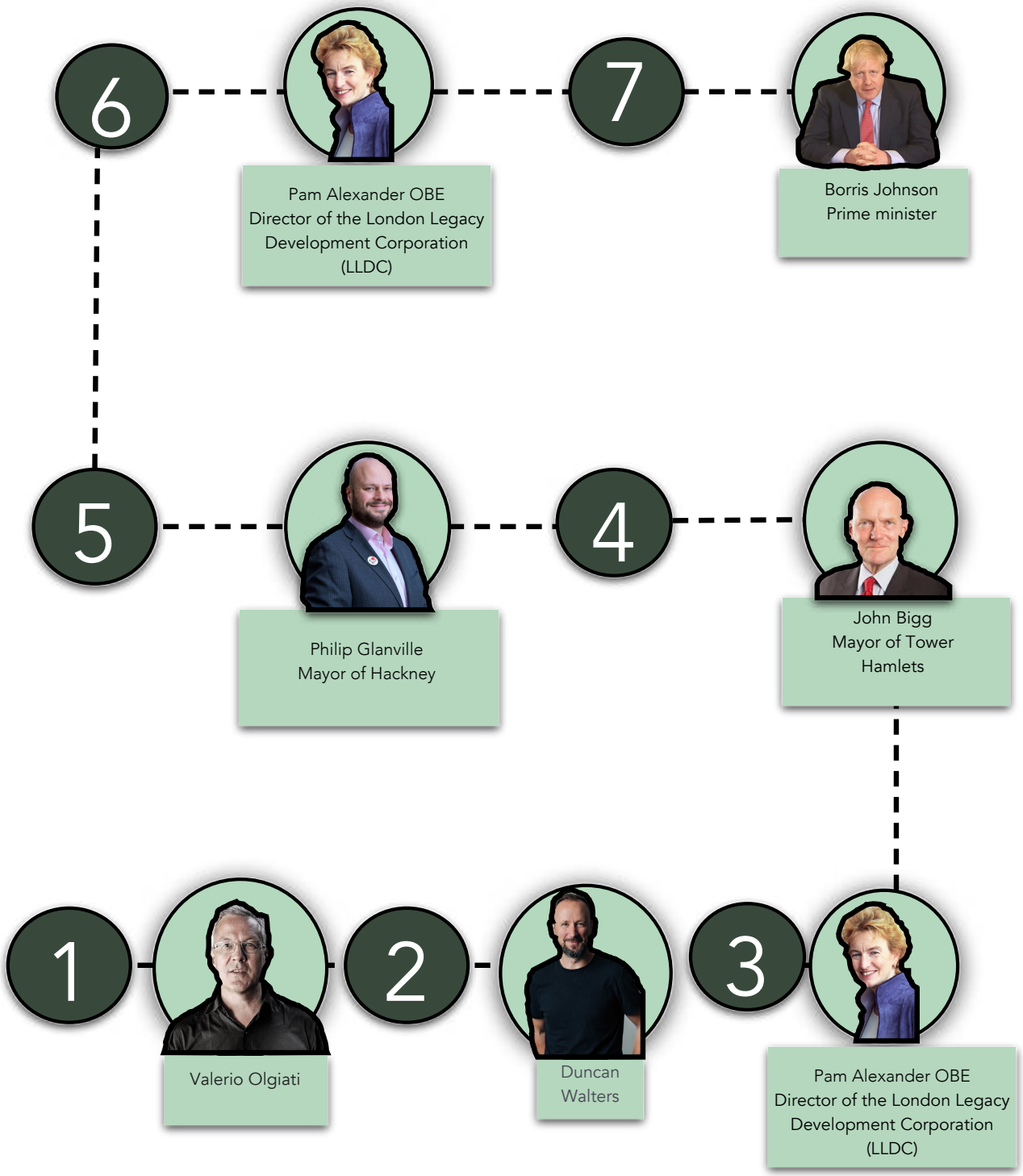




The procurement team must acknowledge the risks involved, which must be considered when designing, as any errors may affect the funding need. If injuries take place, one could file a lawsuit against the procurement team. Unfortunately, some risks are out of one's control, but designing strategies will help to avoid these risks such as risk of falling, security and privacy of the hemp growing and designing on a slope.

Strategic decisions to fix such problems are needed. Strategies such as rails, shutter system and placing the grow bed along the slope till the ground levels are ways to minimise risk. Highlighted in the Constructive design regulations, the strategies are key to ensure no one is hurt or harmed in any way whatsoever.

Stakeholders



The stakeholder profile is essential in procuring the funding. To get the funds requested, all involved must be on board. The design will not only benefit the environment but the surrounding area. As the design aims to be off-grid, the costs will be high to ensure the correct technology is procured. If the design does not use the energy of the surrounding area, less pressure will be put on the surrounding areas, and in the long term, the government will be saving money.

The nature of design requires a Industrial Hemp (low THC Cannabis) cultivation licence and highly expensive technology thus the funding will hopefully be procured from the government with Boris Johnson giving the final decision on whether or not the design will be given the go ahead. Pam, as stated in the project team, will be the key to opening equal doors for all.

Funding & Costs

		Quantity	Price	Total
	Timber	2000	5	10000
	Gutter	2000	4.71	9420
	Nylon	360	8.59	3092.4
	Hemp seeds(Felina 32)	5000 per 1	45	225
	Net pots	5500	1.65	9075
	PVC pipe	100	4.71	104.71
	Hemp growing licence	1	580	580
	Solar panels	60	281.58	16894.8
	Powerwall	150	8401	1260000
	Hemp (180l)	2500	9	22500
	Binder (18kg)	2500	14.5	36250
	Timber	1000	5	5000
	Stone(FLAT COPING (L) 600mm x (W) 400r	1000	14	14000
	Double glaze glass 2000mmx1000mm	100	210	21000
			total	1408142

The funding needed to produce an off-gird building in the United Kingdom is high due to the low sun hours in the colder months. To provide the required energy, the design requires expensive technology supplied by Tesla, as their technology is new, the price is high with the cost of £8400.00 per Powerwall. The price is driven up to an extent where it may seem unfeasible and unnecessary. However, if granted, the design will be able to run solely off the power walls stored electricity for 3 months. This aspect means less power is needed in the area and in the long-term money will be saved due to lowered energy demands

The sum of £1 408 142.00 is requested and may be adjusted per the stakeholder's opinions or suggestions. This price is negotiable and can be lowered with fewer power walls being bought.

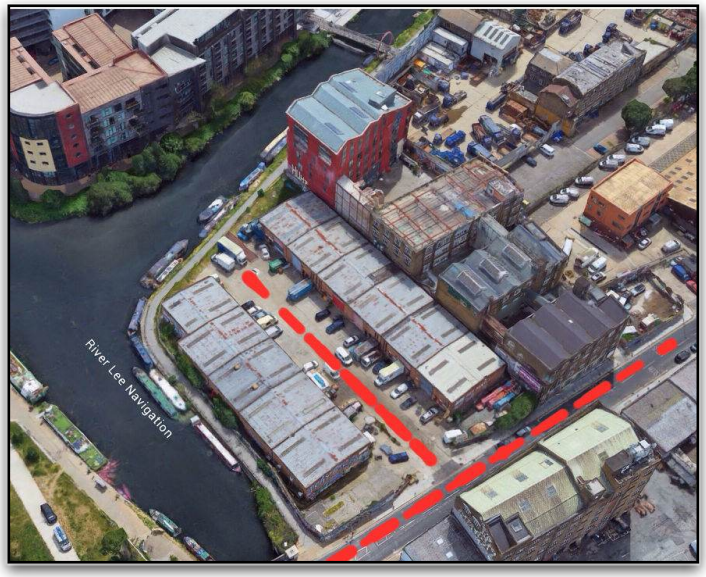
Opportunities and constraints

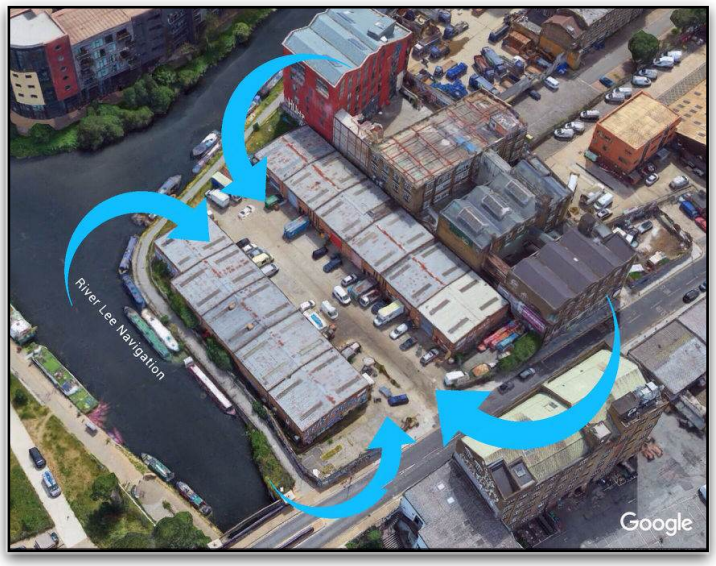
Opportunities





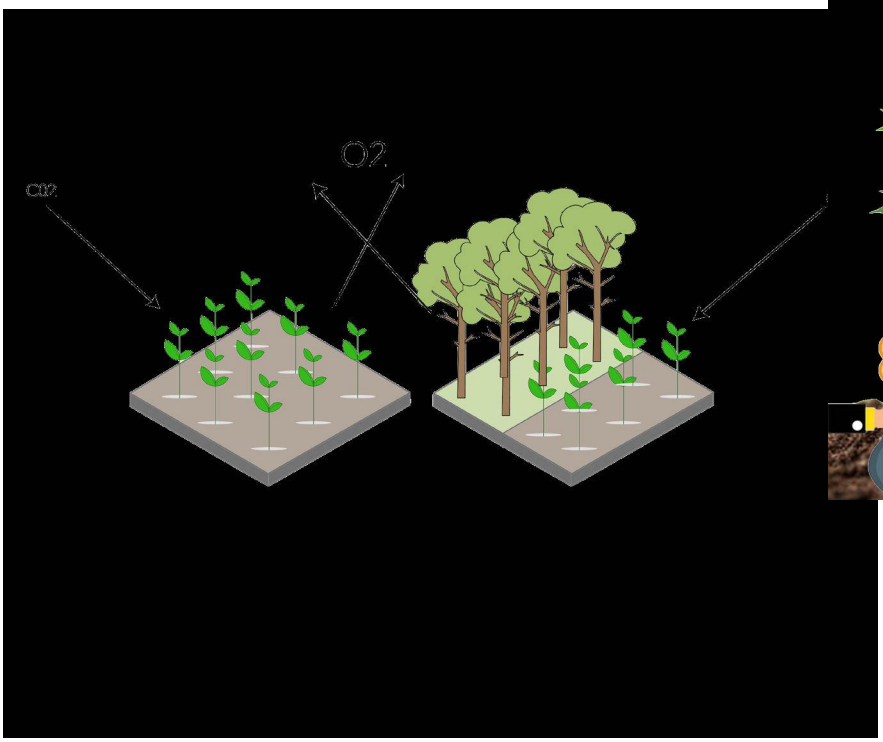
Constraints





Access to the river allows for the water to be used and recycled when needed.	One road access means cars can only enter the site from one direction and a path is needed to ensure the social housing can be built as well as parking for
There is an underground tube station close by and public access to the site is a 5 minute walk from major public transport links	Varied wind directions may cause issues with constructing and dust may travel uncontrollably during the construction.

Value





The value of the project is both to the people of Hackney Wick, Fish Island and the environment.

The growth and production of plants create healthier environments for all living things, which increases social wealth due to improved air quality from the growth cycle of plants. Social-welfare improvement due to job creation and improved environmental benefits is the result of jobs created from the plant processing; a renewable, sustainable economy can be created. A solution for today, tomorrow, and forever on. Plants as a structural and material system to keep the lights on. Benefiting the people is one of the values brought about when creating an urban hemp farm. The second value is to the environment of mother nature."During the growth period "hemp can sequest 15 tons of CO2 per hectare" stated by (Wilson, 2016, p. 2), further backed by (Tradical, 2019)) which explain that one hectare can sequest "18 tons of CO2". When processed, it picks up 10 to 11 tons leaving 4 to 7 tons being absorbed.

Conclusion

A strategy to procure a team and funding to change a community. The process explained detailed procedures that if done correctly, may lead to an off-grid urban hemp farm that combines many people into one community. Back to the basics of past processing and ways of cultivating to produce a low energy demand building drawing people in through form and function. An agricultural landmark that creates an escape from the city to a place to serve new ways of living. Multi-use in nature but working with nature, promoting jobs and working with all. A grow-process-make-build horizon to light new pathways of creating the built environment using both passive and active strategies to further lower energy demand. Functioning as a design to collaborate with people to develop a circular economy where everyone plays a role to form one systematic way of working that allows all parties to benefit.

Moving forward, further research is required into the total energy demand of the building to gather a more precise price calculation as well as active strategies to lower energy demands that are lower in price.

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2023

This report sets out the strategic, legislative and procurement pathway for Back to Basics, an off-grid hemp farm and housing scheme in Hackney Wick. It concludes that a phased, community-led procurement strategy, combined with passive and active energy strategies, can deliver a low-energy, circular development, though further research into total energy demand and pricing is recommended.



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